PREDICT

WEEK 8

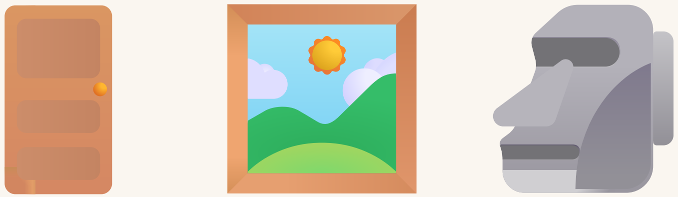
My Own Museum

Final project — everything together

마지막 프로젝트 — 배운 모든 걸 하나로

BIG PICTURE

A room visitors walk into 🏛️



Flat pictures (2D) + solid sculptures (3D) in one gallery.

2D 그림과 3D 조각을 한 전시실에

HOW TO COUNT

Count the axis — one block, one number

1. Stand on the start block — that spot is `(0, 0, 0)`.
2. **x** → walk **right**. Add **+1** for every block.
3. **y** → jump **up**. Add **+1** for every block.
4. **z** → walk **forward**. Add **+1** for every block.

Say them in order every time: `(x, y, z)` = **across, up, forward**.

한 칸 움직이면 +1 — 오른쪽 x, 위로 y, 앞으로 z. 순서는 늘 (x, y, z)

TYPE IT IN

Put the numbers into Minecraft ⌨️

x	y	z
across →	up ↑	forward ⤴

```
place STONE at (x, y, z)
```

- ✓ Open **Code Builder** (press `C` in the world)
- ✓ Type the numbers in order — `(x, y, z)`
- ✓ Press **Run** ▶ — your block appears at that spot

코드 빌더(C) 열고 → (x, y, z) 순서로 숫자 넣고 → Run ▶

CONCEPT

2D vs 3D art

2D picture

`flat on the wall`
`only x & y change`

3D sculpture

`solid in the room`
`x, y, z all used`

그림은 평평(x,y), 조각은 입체(x,y,z)

CONCEPT

The whole course in one build

- **Loops** draw the pictures, row by row
- **fill** makes the room hollow + carves the doorway
- **fill** stacks the pedestals
- **(x, y, z)** places every sculpture on its stand

반복문 · fill · 좌표 — 전부 한 번에

CODE

2D wall picture (z stays 0)

```
for x in range(5):
    blocks.place(RED_WOOL, pos(x, 1, 0))
```

z stays `0`, so it's **flat** on the back wall. Only x changes along the row.

z를 0으로 고정 → 벽에 평평하게

CODE

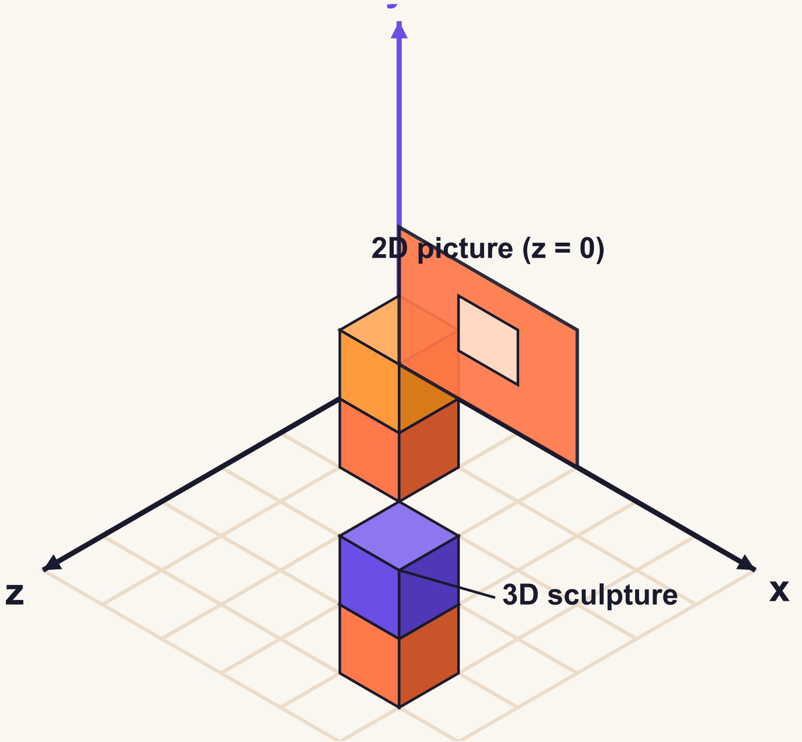
3D sculpture on a pedestal

```
blocks.fill(STONE, pos(4,0,4), pos(5,1,5), FillOperation.REPLACE)
blocks.place(GOLD_BLOCK, pos(4, 2, 4))
blocks.place(GOLD_BLOCK, pos(5, 2, 5))
```

All three numbers used — visitors can walk all the way around it.

3D SPACE

2D pictures + 3D sculptures together



벽에 붙은 평평한 그림(z=0)과 바닥에 선 입체 조각 — 2D와 3D를 한 장면에

PREDICT 🤔

Flat or solid?

Predict first — then click reveal

```
set x to 0
repeat 5 times:
    place red block at (x, 1, 0)
    add 1 to x
```

Reveal output 🔍 reveal

VOCAB

New words today

`museum / gallery`	the room that holds the artworks	미술관 / 전시실
`2D art`	flat — only x and y change (z fixed)	2차원 작품
`3D art`	solid — uses x, y, and z	3차원 작품
`footprint`	the (x, z) space a piece takes on the floor	바닥 자리

🐛 DEBUG

What's wrong with this code?

Goal: a wall picture that climbs **up** the wall (z stays 0).

buggy

```
repeat 4 times: place red at (x, 1, 0)
repeat 4 times: place red at (x, 1, 1)
```

The second row should be higher, not forward. Think first.

DEBUG

The bug

Bug: the second row raises **z** (steps forward onto the floor). To climb the wall, raise **y** instead and keep z = 0.

fixed

```
repeat 4 times: place red at (x, 1, 0)
repeat 4 times: place red at (x, 2, 0)
```

YOUR TURN!

Build your museum 🏛️

- ✓ A gallery room with floor, walls, and a doorway
- ✓ Two flat **2D** pictures (only x & y change)
- ✓ Two **3D** sculptures, each on its own pedestal
- ✓ Pedestals don't overlap; statues rest on top
- ✓ Record a full gallery tour video

QUIZ

What makes a wall picture **2D** (flat)?

- It uses fill instead of loops.
- It's made of gold blocks.
- z stays fixed; only x and y change.
- All three numbers change.

QUIZ

What makes a sculpture **3D** (solid)?

- It floats in the air.
- It uses all three numbers — x, y, and z.
- It only uses x and y.
- It's flat on the wall.

📝 QUIZ

QUESTION 3 OF 3

Your gallery is a solid stone block. What step did you skip?

- Hollowing the inside with fill air.
- Nothing — it's done.
- Adding more pictures.
- Making it taller.

QUIZ

Your score

0 / 3

📖 Re-read the slides. You'll get it.

↺ retake

🏛️

Museum complete!

You can place a block anywhere in space. 🎉

이제 공간 어디든 블록을 놓을 수 있어요